DGMS JSO GEOLOGY 2019 • GEOCHEMICAL EXPLORATION & ORE PROSPECTING

Question 100: Geochemical Anomalies in Mineral Exploration

Statistical Definitions of Background, Threshold, and Primary/Secondary Dispersion Halos

Q.100: Which one of the following can be termed as significant geochemical anomaly in mineral exploration?

(a) Concentration of indicator elements lower than the local threshold

(b) Concentration of indicator elements falling within the regional threshold

(c) Concentration of indicator elements falling within the local threshold

✓ (d) Concentration of indicator elements higher than the local background

✓ Correct Option: (d) Concentration of indicator elements higher than the local background

1. Executive Summary & Fundamental Exploration Axiom

In applied exploration geochemistry (Hawkes & Webb, 1962; Rose, Hawkes & Webb, 1979; Levinson, 1980), the ultimate objective of soil, stream-sediment, and lithogeochemical surveys is to detect concealed mineral deposits by distinguishing between the natural barren geochemical variability of barren rock and abnormal chemical accumulations produced by ore-forming hydrothermal or supergene systems.

A **significant geochemical anomaly** is defined as a statistically and geologically meaningful departure from the normal baseline, characterized by a **concentration of indicator or**

pathfinder elements substantially higher than the local background and exceeding the upper cutoff known as the **geochemical threshold**.

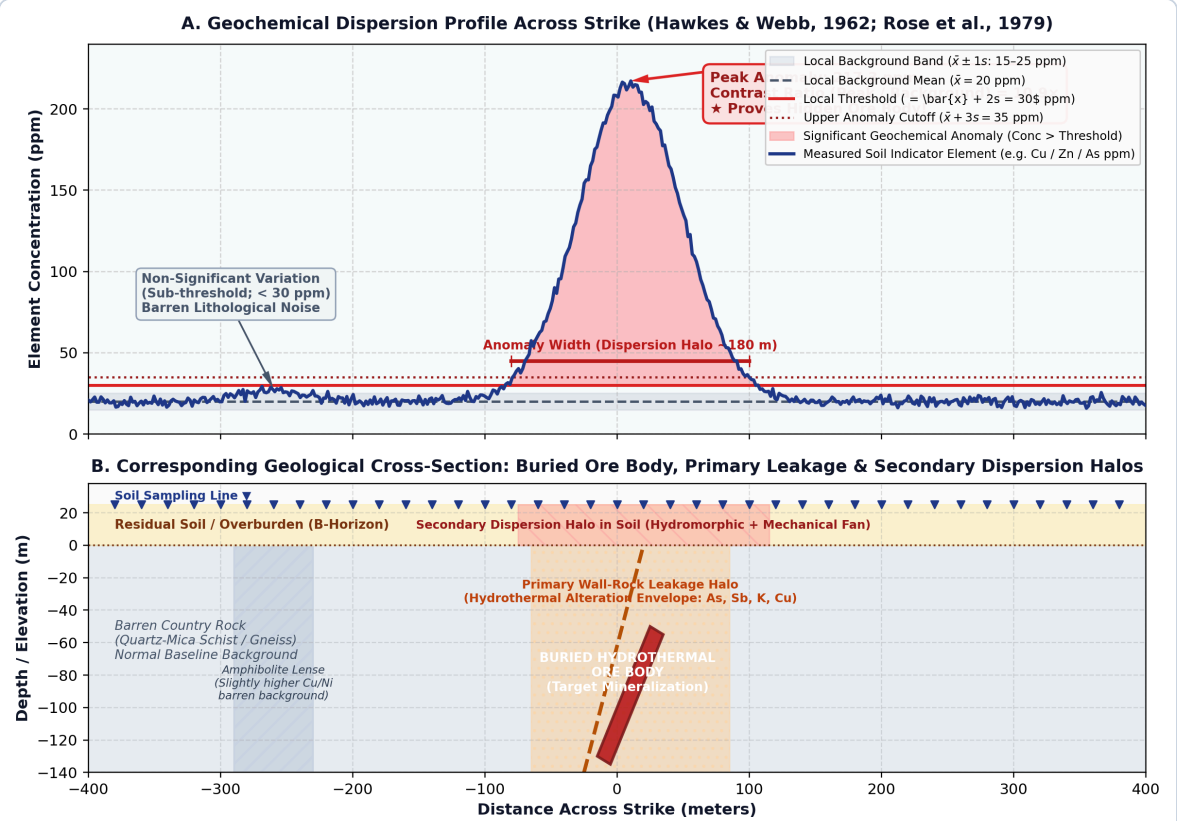


Figure 100.1: Definitive two-panel geochemical exploration model. **(A) Surface Soil Dispersion Profile:** Cross-strike traverse demonstrating the baseline Local Background ($\bar{x} = 20\text{ ppm}$), normal background band ($\bar{x} \pm 1s$: 15–25 ppm), and the statistical Local Threshold ($T = \bar{x} + 2s = 30\text{ ppm}$). Notice that minor sub-threshold lithological noise (left peak at 28 ppm) is non-significant, whereas the peak directly above the ore body reaches 220 ppm (Contrast Ratio $11\times$), delineating a *Significant Geochemical Anomaly*. **(B) Geological Cross-Section:** Subsurface architecture showing the blind hydrothermal sulfide ore shoot, steep feeder shear zone, primary wall-rock leakage alteration envelope, and the resultant secondary mechanical/hydromorphic dispersion fan in residual B-horizon soils sampled at regular survey stations.

2. The Triad of Exploration Geochemistry: Background, Threshold & Anomaly

Every geochemical interpretation rests upon the rigorous statistical and geological definition of three interdependent parameters:

A. Geochemical Background ($\bar{x}$ or b)

- **Definition:** The normal abundance or concentration range of an element in unmineralized, barren earth materials (rocks, soils, glacial tills, or stream sediments) within a specific geological terrane.
- **Ahrens' Law of Lognormal Distribution (1954):** Trace elements in geological media rarely follow a standard Gaussian normal distribution; rather, they almost universally obey a **lognormal distribution**. Consequently, geochemical datasets are routinely transformed to logarithms ($\log_{10} x$) prior to calculating parametric statistics.
- **Local vs. Regional Background:** *Regional background* reflects broad crustal Clarke abundances across an entire metallogenic province. However, exploration requires evaluating the **Local Background**, which is calibrated strictly against the specific host lithology within the immediate survey area (e.g., ultramafic rocks inherently possess a high Ni-Cr background, whereas granites possess low Ni-Cr but high U-Th-Rb background).

B. Geochemical Threshold (T)

- **Definition:** The upper limit of normal background variation, serving as the critical boundary line that separates barren baseline noise from potentially anomalous samples.
- **Parametric Formulation (Hawkes & Webb, 1962):**

  > Parametric Threshold (T) = $\bar{x} + 2s$
  > (Where $\bar{x}$ = mean background, and s = standard deviation)

  Assuming a normal distribution (or log-transformed distribution), approximately **95.45%** of barren samples lie within $\bar{x} \pm 2s$. In a one-tailed exploration test, only **2.28%** of unmineralized samples will by random chance exceed $\bar{x} + 2s$.
- **High-Confidence Threshold (T_{99}):** For high-risk, expensive drilling campaigns, geochemists often set $T = \bar{x} + 3s$, where only **0.13%** of barren samples exceed this cutoff.
- **Non-Parametric & EDA Thresholds (Tukey, 1977; Sinclair, 1974):** To prevent extreme outlier ore samples from artificially inflating $\bar{x}$ and s, robust non-parametric methods are widely adopted:

  > Median Absolute Deviation (MAD): $T = \text{Median} + 2 \times \text{MAD}$

> Tukey Boxplot Upper Fence: $T = Q_3 + 1.5 \times \text{IQR}$ (where $\text{IQR} = Q_3 - Q_1$)

C. Geochemical Anomaly (A)

- **Definition:** Any marked deviation from the geochemical background. In mineral prospecting, a **significant anomaly** is an element concentration that is **substantially higher than the local background** ($A > T > \bar{x}_{\text{local}}$) and genetically related to primary or secondary dispersion halos of an ore deposit.
- **Anomaly Contrast Ratio (C):**

> Contrast Ratio (C) = $\frac{\text{Peak Anomaly Concentration}}{\text{Local Background Mean (\bar{x})}}$

High-contrast anomalies ($C \ge 5\times$ to $20\times$) indicate high prospectivity, whereas low-contrast anomalies ($C < 2\times$) require multi-element pathfinder verification.

3. Indicator Elements vs. Pathfinder Elements

Exploration programs utilize two complementary suites of geochemical elements to delineate targets:

- **Indicator (Target) Elements:** The primary metal or commodity sought for commercial mining (e.g., Au, Cu, Pb, Zn, Ni, Sn, U). In many deposits (such as gold), the indicator element may be immobile, coarse-grained ("nugget effect"), or low in abundance.
- **Pathfinder (Pioneer) Elements:** Mobile, volatile, or highly soluble trace elements associated with the ore-forming fluid that form broader, more uniform primary or secondary halos, providing a much larger geochemical target for regional exploration.

Deposit Type	Primary Target (Indicator Element)	Diagnostic Pathfinder Elements	Geochemical Rationale
Orogenic Gold Quartz Veins	Au	$\text{As, Sb, Bi, Te, W, B}$	Arsenic and antimony form expansive volatile alteration halos around gold-bearing shear zones.

Deposit Type	Primary Target (Indicator Element)	Diagnostic Pathfinder Elements	Geochemical Rationale
Epithermal Precious Metals (Au-Ag)	Au, Ag	$\text{Hg, As, Sb, Tl, Ba, F}$	Low-temperature volatile metals (Hg, Tl) migrate to shallow, near-surface levels above boiling zones.
Porphyry Copper-Molybdenum	Cu, Mo	$\text{Re, Au, As, Zn, Pb, Se, Te}$	Concentric geochemical zoning: central $\text{Cu-Mo} \rightarrow$ peripheral $\text{Pb-Zn} \rightarrow$ distal As-Sb-Mn.
Volcanogenic Massive Sulfides (VMS)	Cu, Zn, Pb	$\text{Ba, Ag, As, Cd, Au, Se}$	Barium forms extensive hydrothermal barite caps; Cd substitutes for zinc in sphalerite.
Uranium Unconformity / Roll-Front	U	$\text{Ra, Rn, He, V, Mo, Se, As}$	Gaseous radon ($\text{}^{222}\text{Rn}$) and helium migrate through fractures; selenium marks the redox front.
Rare Metal Pegmatites (LCT)	Li, Ta, Cs	$\text{Rb, Be, Sn, Nb, B, F, Ga}$	Highly incompatible elements disperse into exocontact metasomatic halos in host amphibolites/schists.

4. Primary vs. Secondary Geochemical Dispersion Halos

Geochemical anomalies are generated by two sequential dispersion mechanisms:

- **Primary Dispersion Halos (Syngenetic / Epigenetic):** Formed during mineralization and wall-rock hydrothermal alteration. Halos exhibit chemical zoning controlled by temperature-pressure gradients, fluid phase separation, and fluid-rock interaction (e.g., potassic, phyllic, and argillic alteration envelopes).
- **Secondary Dispersion Halos (Supergene / Surficial):** Formed when bedrock deposits undergo weathering, pedogenesis, and erosion. Metals disperse mechanically (gravity, colluvium, stream sediment detritus) or hydromorphically (dissolved ions in

groundwater and soil water controlled by pH-Eh, adsorption on Fe-Mn oxyhydroxides, and organic complexation).

5. Significant Anomalies vs. Non-Significant (False / Spurious) Anomalies

A critical competency tested in examination geology is recognizing that **not every elevated concentration constitutes an economic discovery**:

- **Significant (True) Geochemical Anomaly:** Arises directly from primary hydrothermal ore mineralization or its weathered secondary dispersion halo. It warrants capital investment, trenching, induced polarization (IP) geophysics, and drilling.
- **Non-Significant (Spurious / False) Geochemical Anomaly:** Caused by non-economic geological or environmental factors:
  1. *Lithological Variation:* Normal unmineralized serpentinite containing 2,000 ppm Ni and Cr adjacent to sandstone containing 10 ppm Ni. The high values in serpentinite represent barren local background, not an ore anomaly!
  2. *Scavenger Effect:* Precipitation of amorphous iron-manganese oxides in stream seepages strongly adsorbs trace elements (Co, Ni, Cu, Zn, U), generating pseudo-anomalies in barren drainage basins.
  3. *Organic Enrichment:* Peat bogs and organic-rich swamp muck chelate uranium and copper from dilute groundwaters.
  4. *Anthropogenic Contamination:* Smelter smoke plume fallout, old tailing dumps, mining roads paved with waste rock, or agricultural fertilizers enriched in cadmium and phosphorus.

6. Detailed Option Elimination Analysis

Option	Concentration Condition	Geochemical Validity & Exploration Significance	Status
(a)	Concentration of indicator elements lower than the local threshold	**Incorrect.** Values below the threshold ($C < T$) represent ordinary, barren background variability or zones of geochemical depletion. They do not constitute an anomaly.	**Eliminated**

Option	Concentration Condition	Geochemical Validity & Exploration Significance	Status
(b)	Concentration of indicator elements falling within the regional threshold	**Incorrect.** A regional threshold is calibrated across vast, diverse geological provinces. Falling *within* the regional threshold signifies broad crustal baseline fluctuations, completely failing to isolate localized mineralizing systems.	**Eliminated**
(c)	Concentration of indicator elements falling within the local threshold	**Incorrect.** The local threshold ($T = \bar{x} + 2s$) is the upper boundary of normal unmineralized background. Values falling *within* or below this threshold represent expected statistical noise in barren rock.	**Eliminated**
(d)	**Concentration of indicator elements higher than the local background**	**CORRECT.** By classical geochemical definition, an exploration anomaly is an abnormal concentration of target or pathfinder indicator elements that rises **substantially higher than the local background** and surpasses the local threshold ($A > T > \bar{x}_{\text{local}}$), indicating proximity to economic mineralization.	✓ **CORRECT**

7. Step-by-Step Worked Numerical Exploration Example

Field Exploration Case Study: Soil Traverse over a Concealed Copper Lode

An exploration company samples B-horizon soil along a 10-station traverse across a quartz-chlorite schist terrane. Copper concentrations (Cu ppm) are recorded as follows:

> Stations 1 to 8 (Barren Background Rocks): 18, 22, 19, 25, 21, 23, 20, 24 ppm
> Station 9 (Over Altered Fracture Zone): 58 ppm
> Station 10 (Directly Over Vein Apex): 185 ppm

Step 1: Calculate Local Background Mean ($\bar{x}$) and Standard Deviation (s) of Barren Host Rock:

- Sum of barren stations (1 to 8) = 172 ppm • Sample count $n = 8$
- $\bar{x}_{\text{background}} = \frac{172}{8} = \mathbf{21.5\text{ ppm}}$
- Sample standard deviation $s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} = \mathbf{2.45\text{ ppm}}$

Step 2: Determine Local Geochemical Threshold (T):

- $T = \bar{x} + 2s = 21.5 + 2(2.45) = 21.5 + 4.9 = \mathbf{26.4\text{ ppm}}$

Step 3: Classify Stations:

- Stations 1–8 ($18–25\text{ ppm}$): All values are $\le 26.4\text{ ppm}$ → *Normal Local Background (Barren).*
- Station 9 (58 ppm): Substantially higher than background ($58 > 26.4\text{ ppm}$, Contrast = $2.7\times$) → **Significant Geochemical Anomaly (Secondary Dispersion Halo).**
- Station 10 (185 ppm): Extremely elevated ($185 > 26.4\text{ ppm}$, Contrast = $8.6\times$) → **Peak Significant Geochemical Anomaly (Drill Target).**

> **Key Examination Takeaway**
>
> Whenever an examination question presents background and threshold options, remember: **Values falling within or below thresholds are ordinary barren background noise.** An anomaly is strictly defined by an accumulation of indicator elements **higher than the local background** that breaks above the local threshold cutoff.

8. Standard Literature & Authoritative References

- **H. E. Hawkes & J. S. Webb** (1962), *Geochemistry in Mineral Exploration*, Harper & Row, New York, Chapter 2: "Principles of Geochemical Dispersion" and Chapter 3: "Background and Anomalies", pp. 23–58.

- **A. W. Rose, H. E. Hawkes, & J. S. Webb** (1979), *Geochemistry in Mineral Exploration* (2nd Edition), Academic Press, London, Chapter 3: "Statistical Interpretation of Geochemical Data", pp. 31–64.
- **A. A. Levinson** (1980), *Introduction to Exploration Geochemistry* (2nd Edition), Applied Publishing Ltd., Calgary, Chapter 2: "Primary Dispersion" and Chapter 6: "The Geochemical Anomaly", pp. 195–242.
- **A. J. Sinclair** (1974), *Selection of Threshold Values in Geochemical Data Using Probability Graphs*, Journal of Geochemical Exploration, Vol. 3, pp. 129–149.
- **G. J. S. Govett** (1983), *Handbook of Exploration Geochemistry: Rock Geochemistry in Mineral Exploration* (Vol. 3), Elsevier Scientific Publishing, Amsterdam, pp. 45–88.